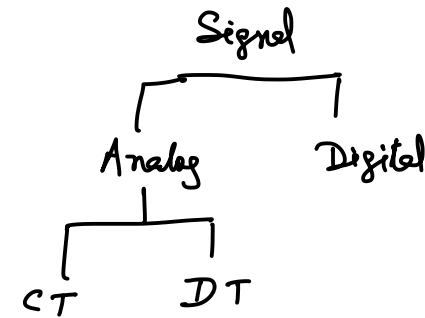


Why and Whynot DSP?

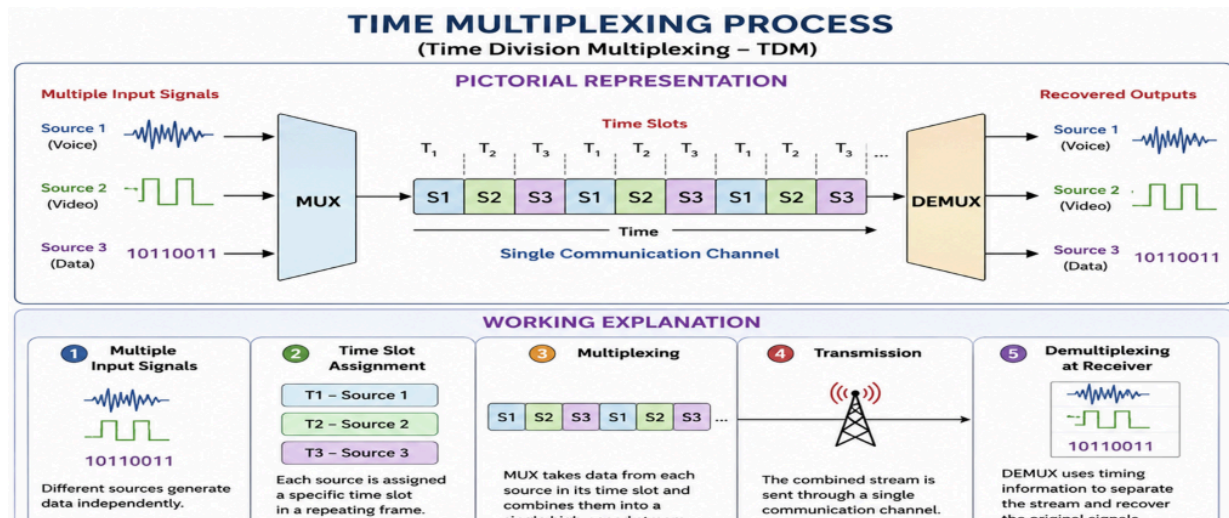
Friday, July 24, 2026 8:30 AM

Why?

- Minimal sensitivity to component tolerance and environmental changes (Noise)
- Volume production without adjustment
- Digital circuits are fully integrable
- With word-length (bits) DSP accuracy increases and also COST (More bits → more accurate value)
- Dynamic range of DSP is almost unlimited
- DSP system can perform multiple tasks together unlike ASP
- DSP processors characteristics is adjustable. (E.g., ADAPTIVE FILTER)
- DSP provide exact linear phase (No delay distortion) system
- DSP provide multirate signal processing
- DSP provide no issue of loading problem due to cascading
- Digital signal can easily be stored in magnetic tap for a long time
- DSP can process very low frequency signal without issues



$$DR = \frac{\text{Strongest value}}{\text{Weakest value}}$$





Why not?

- Increased complexity (e.g., different components in DSP framework)
- Frequency range in operating DSP is limited (Sampling rate is limited)
- Power dissipation

Signal Representation

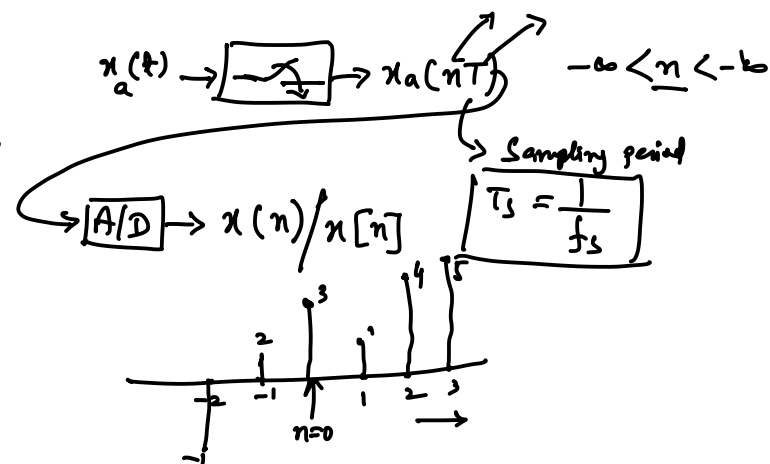
Sunday, July 26, 2026 10:53 PM

- Sequential, Tabular, Graphical,
- Real and Complex signal

$$x(n) = \begin{cases} 1, 2, 3, 4, 5, 6 \\ -1, 2, 3, 1, 4, 5 \end{cases}$$

$\uparrow$
 $n=0$
 $n=-2, n=-1, n=1, n=2, n=3$

$$x(n) = \begin{cases} n, & 0 < n < 10 \\ -n, & \text{elsewhere} \end{cases}$$



$$x(n) = \{1, 2, 3, 4, 5\}$$

$\uparrow$
 $n=0$
 $n=-1, n=1, n=2, n=3$

$$= \{ -1+j, j, j+2, +3j \}$$

$\uparrow$
 $j+2$

Operations on Signal

Sunday, July 26, 2026 11:09 PM

- Addition/Subtraction, Multiplication
- Folding and Shifting
- Some Interesting Facts



$$x(n) \xrightarrow{\sum} z(n) = x(n) + y(n)$$

$$x(n) = [1, 2, 3, 4, 0] \quad y(n) = [-1, 0, 1, 3]$$

$$z(n) = [1, 1, 3, 5, 3]$$

$$z(n) = x(n) \pm y(n)$$

$$z(n) = x(n) * y(n) = [0, -2, 0, 4, 0]$$



Delay / Advance

Time Reversal / Folding

$$x(n) = x(-n)$$

$$x(n) = \{1, 2, 0, 3, 4\}$$

$$x(-n) = \{4, 3, 0, 2, 1\}$$

$$x(n) = [-1, 2, 1, 3, -2]$$

$$\left. \begin{aligned} x(n-1) &= \text{Delay} \\ x(n+1) &= \text{Advance} \end{aligned} \right\}$$

$$x(n) \xrightarrow{D/A} x(n-k)$$

$$k = +ve$$

$$k = -ve$$

$$x(n-1) = [-1, 2, 1, 3, -2]$$

$$x(n) \rightarrow x(-n-k)$$

$$x(n+1) = [-1, 2, 1, 3, -2]$$

$$x(n) = \{-1, 2, 1, 3, -2\}$$

$$x(n-2) = \{-1, 2, 1, 3, -2\}$$

$$x(n-3) = \{0, -1, 2, 1, 3, -2\}$$

$$x(n) = \{-1, 4, 1, 3, -2\}$$

$$x(-n) = \{-2, 3, 1, 4, -1\}$$

$$\underline{x(-n+1)} = \{-2, 3, 1, 4, -1\} \checkmark$$

$$\underline{x(-n-1)} = \{-2, 3, 1, 4, -1\} \checkmark$$

$$x(n) = \{-1, 2, 0, -2, 3\}$$

$n \Rightarrow$

-3, -2, -1, 0, 1, 2, 3

$$x(n) = x(-3), x(-2), x(-1), x(0), x(1), x(2), x(3)$$

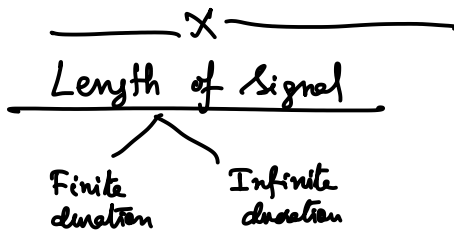
$$x(-n) \quad \checkmark \quad \checkmark \quad \checkmark \quad \checkmark \quad \checkmark \quad \checkmark$$

$$\underline{x(n-1)} \quad \checkmark \quad \checkmark \quad \checkmark \quad \checkmark \quad \checkmark \quad \checkmark$$

$$\underline{x(n-2)} \quad \checkmark \quad \checkmark \quad \checkmark \quad \checkmark \quad \checkmark \quad \checkmark$$

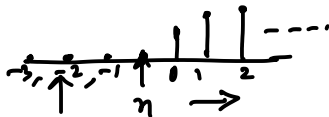
$$x(-n+1) \quad \checkmark \quad \checkmark \quad \checkmark \quad \checkmark \quad \checkmark \quad \checkmark$$

$$x(-n-1) \quad \checkmark \quad \checkmark \quad \checkmark \quad \checkmark \quad \checkmark \quad \checkmark$$



$$x(n) = \begin{cases} 0 < n < K & \text{Finite} \\ -\infty < n < \infty & \rightarrow \text{Infinite} \end{cases}$$

$$\underline{-K < n < K}$$



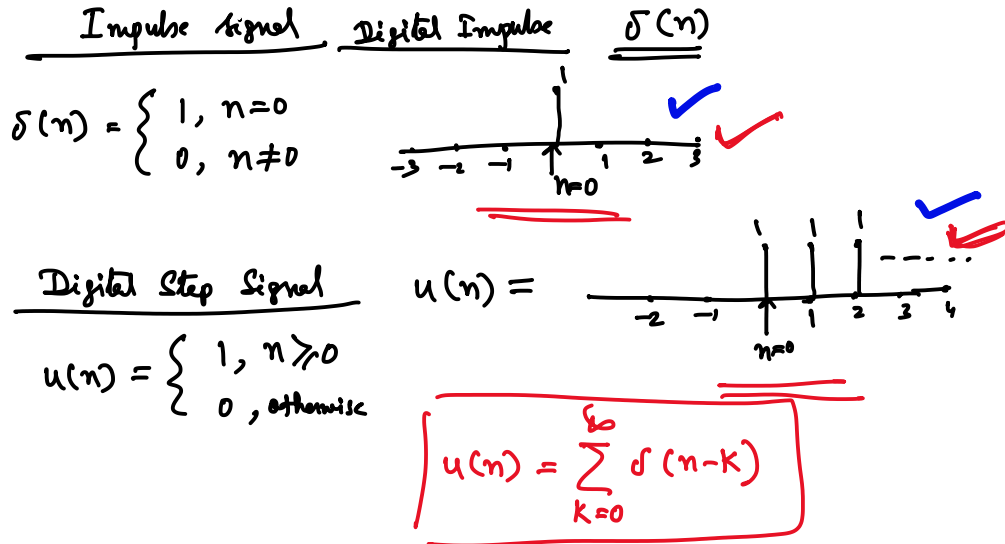
$$x(n) = \{ \underset{\uparrow}{0}, 1, 2, 3, \dots \}$$

$$y(n) = \{ \dots, 1, 2, \underset{\uparrow}{3} \}$$

Elementary Signal

Sunday, July 26, 2026 11:14 PM

- Impulse, Step
- $\delta(n)$ using $u(n)$, and vice-versa
- Expressing any arbitrary signal with $\delta(n)$, $u(n)$



$$\delta(n) = u(n) \cdot [u(-n)]$$

$$\delta(n) = u(n) - u(n-1)$$

$$1) \underbrace{u(n) + u(-n)}_{z(n)} = 1 + \delta(n) \quad \forall n$$

$$u(n) = \{1, 1, 1, \dots\} \checkmark$$

$$u(-n) = \{\dots, 1, 1, 1\} \checkmark$$

$$n=0, \quad z(n) = 2$$

$$n=1, \quad z(n) = 1$$

$$n=2, \quad z(n) = 1$$

$$n=-1, \quad z(n) = 1$$

$$n=-2, \quad z(n) = 1$$

$$2) \boxed{u(n) + u(-n-1) = 1} \quad \forall n$$

$$3) u(n) - u(-n) = ?$$

$$4) u(n) - u(-n-1) = ?$$

Even/Odd

Sunday, July 26, 2026 11:00 PM

Even Signal, Odd Signal, Signal Decomposition

↓
Symmetrical

↑
non-symmetrical

$$x(n) = \{1, 2, 3, 3, 4, 3\}$$

$$x(n) = \{1, 2, 3, 4, 3, 2, 1\}$$

$$x_1(n) = \{1, 2, 3, 1, 1, 2, 3\}$$

$$x_e(n) = x_e(-n)$$

↓ Even

$$\text{odd} \Rightarrow x_o(n) = -x_o(-n)$$

$$\Rightarrow x(n) = \{4, -3, -2, 1, 2, 3, 4\}$$

$$x_m(n) = x_e(n) + x_o(n)$$

$$x_m(n) = \{1, 2, 0, 1, 2, 3\}$$

$$x_m(n) = x_e(n) + x_o(n)$$

$$x_m(-n) = x_e(-n) + x_o(-n)$$

$$x_e(n) = \frac{x_m(n) + x_m(-n)}{2}$$

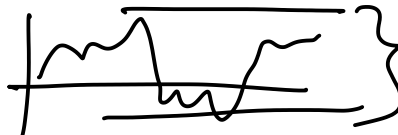
$$x_o(n) = \frac{x_m(n) - x_m(-n)}{2}$$

Periodic, BIBO...

Sunday, July 26, 2026 11:05 PM

- Periodic, BIBO, Absolutely summable,
- Square Summable, Energy, Power (Avg. Power)

Periodic $x(n) = x(n + NK)$ $\rightarrow 1, 2, 3, \dots$
 $N =$
Bounded Signal
 $|x(n)| \leq m < \infty \rightarrow \forall n$
 $\rightarrow$ Any value



1) Absolutely Summable signal $\sum_{n=-\infty}^{\infty} |x(n)| \leq p < \infty$

2) Square Summable Signal $\sum_{n=-\infty}^{\infty} |x(n)|^2 \leq Q < \infty$ $\rightarrow$ SMS
 $\rightarrow$ Finite energy
 $\rightarrow$ Energy Signal

3) Power of the Signal (Average Power) $\Rightarrow$

$$P_{avg} \triangleq \frac{1}{N} \sum_{n=0}^{N-1} |x(n)|^2$$

$$P_{avg} \triangleq \lim_{K \rightarrow \infty} \frac{1}{2K+1} \sum_{n=-K}^K |x(n)|^2$$

Exponential+Sinusoidal

Sunday, July 26, 2026 11:22 PM

- Exponential
- Sinusoidal
- Periodicity of mixed signals

1) Two sided exponential $\rightarrow A\alpha^n = x(n)$

2) one sided exponentials $\rightarrow x(n) = A\alpha^n u(n)$

$\rightarrow$ No if $|\alpha| > 1$
Yes if $|\alpha| \leq 1$

Sinusoidal signal $x(n) = x(n+N)$

$$\Rightarrow \cos(\omega_0 n + \phi) = \cos(\omega_0 (n+N) + \phi)$$

$$\Rightarrow \boxed{\omega_0 N = 2\pi k} \rightarrow \text{Integer multiple of } (2\pi)$$

$$\Rightarrow \boxed{\frac{\omega_0}{2\pi} = \frac{k}{N}}$$

Ex $x(n) = \cos(3n)$

$\cos 3n = \cos \omega_0 n$ $\omega_0 = 3 \Rightarrow \boxed{\frac{\omega_0}{2\pi} = \frac{3}{2\pi}}$

Periodicity of a mixed signal

$x_1(n)$	$x_2(n)$	$x_3(n)$	$x_4(n) = x_1(n) + x_2(n) + x_3(n)$
$\downarrow$	$\downarrow$	$\downarrow$	
$N_1 = 3$	$N_2 = 5$	$N_3 = 12$	

$$\boxed{N_4 = \text{LCM}(N_1, N_2, N_3)} = \underline{\underline{60}}$$

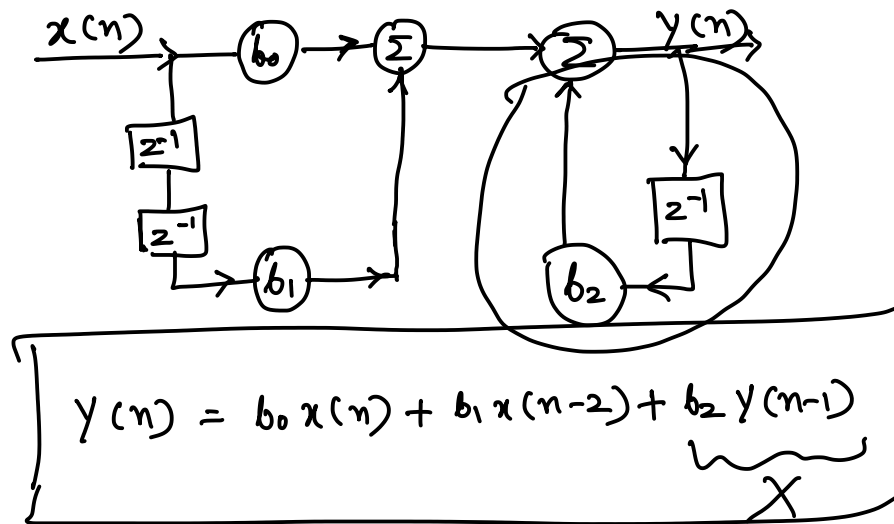
— X —
System

Delay = $\boxed{z^{-1}}$

$$x(n) \rightarrow \boxed{z^{-1}} \rightarrow y(n) = x(n-1)$$

Nth delay $\left[x(n) \rightarrow \boxed{z^{-N}} \rightarrow y(n) \right] \quad \times$

$$x(n) \rightarrow \boxed{z^{-1}} \rightarrow \boxed{z^{-1}} \rightarrow \boxed{z^{-1}} \rightarrow \dots \rightarrow \boxed{z^{-1}} \rightarrow y(n)$$



Expressing Any signal with Elementary signals

Thursday, July 30, 2026 12:42 PM

$$x(n) = \sum_{k=-\infty}^{\infty} x(k) \delta(n-k)$$

$$x(n) = [2, 4, 1] = 2\delta(n) + 4\delta(n-1) + \delta(n-2)$$

$$\uparrow$$

$$= 2u(n) + 2u(n-1) - 3u(n-2) - \underbrace{u(n-3)}_X$$

$$x(n) = \{2, 5, 4, 7\} = 2\delta(n) + 5\delta(n-1) + 4\delta(n-2) + 7\delta(n-3)$$

$$\uparrow$$

$$n=0$$

$$= 2u(n) + 3u(n-1) - u(n-2) + 3u(n-3) - 7u(n-4)$$

$$x(n) = \{0, 1, 2, 3\} = \delta(n-1) + 2\delta(n-2) + 3\delta(n-3)$$